A small-budget retrospective study of Indian mandi price forecasts: reproducible cleaning and dependence sensitivity

C Rishitha, Karthikeya Y, Ashwin Tyagi and Kaushal S

***Abstract*—We report a bounded, clean-source retrospective study of Indian mandi price forecasts, using a fixed historical panel and a small LightGBM search budget. Two cleaning variants are rebuilt with pre- and post-execution source, input and executable hashes. Training, selection and interval calibration use disjoint periods; evaluation covers four horizons in already observed 2026 data. Pooled MAE is 4.4–5.9% below contemporaneous persistence. However, MASE is approximately 1.00–1.11 against training-period naive error scales and omits 4.4–4.5% of rows without usable scales. These different denominators preclude universal superiority claims. Longer circular and non-circular block sensitivities retain the observed advantage, but limited seasons and serial dependence leave inferential coverage unvalidated. Older national-scale and source-completeness results are historical supporting evidence, not reproduced principal findings. A frozen-model issuance recorder supports future collection; outcomes and human statistical review remain pending.**

***Index Terms*—Agricultural prices, Forecast evaluation, Reproducibility, Cleaning sensitivity, Scaled errors, Dependent observations**

I. Introduction

Indian agricultural wholesale markets (*mandis*) report minimum, maximum and modal prices by commodity and variety through AGMARKNET and public data mirrors. These reports provide price information for farmers comparing markets and sale timing, but their value as forecasting data depends on reporting continuity, source completeness and the evaluation protocol. The empirical question is how much a learned forecast improves on simply retaining the latest reported price, and how cautiously that improvement should be interpreted.

The mandi forecasting literature includes studies of individual markets, small groups of markets and monthly indices (Section II), often comparing deep or hybrid models with ARIMA [1]–[3]. Forecast-evaluation research motivates a complementary emphasis: strong simple benchmarks [4], [5], appropriate scaling across heterogeneous series [6], alignment between the training loss and the scoring function [7], [8], and explicit treatment of leakage and data vintage [9], [10]. These considerations motivate an auditable empirical study rather than a claim of model superiority across published studies with different data and protocols.

Our primary contribution is a smaller, resource-limited *clean-source release study* of daily mandi modal-price forecasting. For each horizon and filtering regime, the learner uses 500,000 deterministically sampled training rows from 2024; January–June 2025 is used for model selection and early stopping, July–December 2025 for interval calibration, and January–October 2026 for retrospective evaluation. Targets are purged at period boundaries. The release documents the feature builds and saved predictions, but retains a fixed historical panel as an external input. It therefore supports claims about this release and observed panel, rather than validation of the earlier full national benchmark.

We make three empirical contributions, with their limits stated alongside the findings.

1) **A chronological, baseline-explicit release comparison.** We compare an absolute-loss LightGBM candidate with persistence and four other simple baselines at the next-report, 3-, 7- and 14-day horizons. The two regimes exclude or retain reports flagged by the fivefold jump rule while preserving the other validity checks and panel membership. Across the eight fits, pooled MAE is 4.4–5.9% lower than persistence on the retrospective evaluation rows. These are descriptive historical gains; adoption of LightGBM remains exploratory, with no independent confirmation of its superiority over the earlier Spark models or of benefits to farmer decisions.
2) **Explicit separation of pooled error and scaled error.** Pooled MAE and MASE answer different questions: MASE divides errors by each series' training-period, horizon-specific naive error scale, whereas the pooled MAE comparison uses evaluation-period persistence error as its reference. MASE also omits approximately 4.4–4.5% of forecast rows because their training scales are not positive and finite. It consequently describes a different population from full-row MAE; a MASE value above one is compatible with a pooled MAE gain over evaluation-period persistence.
3) **Dependence-aware sensitivity analysis and interval assessment.** We report calendar-day block-resampling and dependence sensitivities alongside retrospective prediction-interval coverage. Forecast rows are not independent replicates, and the inference relies on approximate weak-dependence and stationarity assumptions conditional on the fitted models and observed panel. Neither small test probabilities nor narrow resampling intervals establish validated inferential coverage or generalisation to independent new markets.

An earlier source-completeness audit provides historical

The authors are with the Department of Artificial Intelligence and Data Science, Amrita Vishwa Vidyapeetham, Coimbatore, India.

Correspondence details require author verification before submission.

supporting evidence that incomplete public mirrors can alter apparent forecast accuracy and interval coverage. That audit is not reproduced as the principal validation of this release, and the release does not establish the original full benchmark's source state. Similarly, the 2026 evaluation is retrospective: its forecasts were not issued as the reports arrived, so it does not constitute prospective confirmation. Independent confirmation, expert statistical review and future-issued evaluation remain outstanding.

II. Related work

a) Forecasting across markets and horizons.: Daily mandi forecasting studies establish the relevance of nonlinear learners but differ substantially in coverage and evaluation. Paul et al. compare ARIMA, GRNN, SVR, random forests and boosting for brinjal in 17 Odisha markets, including predictive-accuracy tests [1]. At broader coverage, monthly prices for 19 commodities in 143 markets favour recurrent models over alternatives including a random walk with drift [11]; daily data for 23 commodities in 165 markets are evaluated with an 80/20 split without MASE [2]. Monthly aggregation, commodity selection and split design prevent direct numerical comparison with daily, target-purged multi-horizon evaluations. Hybrid, wavelet and neural models address individual commodity settings [3], [12]–[17], while combination methods also appear in agricultural forecasting [18]. Model complexity alone is therefore not a research contribution.

b) Evaluation and data provenance.: Forecasting competitions motivate strong simple benchmarks [4], [5], [19]. Scaled errors [6], predictive-accuracy tests [20] and time-series evaluation studies [9], [21], [22] address complementary aspects of evaluation; a small p value cannot replace an effect size or evidence from an independent period. Consistency between training loss and the scored functional matters when many changes are zero [7], [8]. AGMARKNET missingness and erroneous entries have already motivated forecasting frameworks [23], [24]; we do not claim to discover agricultural data-quality problems. Real-time data vintages can also change macroeconomic forecast evaluations [10]. Here the narrower question is whether incomplete mirror coverage changes agricultural evaluation on matched origins when compared with official reports.

c) Uncertainty and decision support.: EnbPI, ensemble conformalised quantile regression and adaptive conformal inference address dependent observations [25]–[27]. Drift-aware conformal prediction has already been applied to Chinese potato prices [28]; our volatility-stratified split-conformal implementation is an evaluated baseline, not a new uncertainty method. Selection and calibration sharing a period in the original experiment motivates the separate-period robustness study below. Information interventions have heterogeneous effects on farmers [29]–[32], so price-prediction gains cannot establish welfare or trading profits.

d) Contribution and scope.: The primary contribution is a bounded, clean-source cleaning and dependence-sensitivity study on a fixed historical panel. It documents a small-budget comparison with five simple baselines, disjoint selection/calibration periods, multiple metric denominators and explicit dependence limitations. This is an empirical execution/evaluation contribution, not a new learner, novel conformal method or claim of unprecedented combination. The earlier matched-origin source-completeness audit is historical supporting motivation, not the principal reproduced validation. Model adoption remains exploratory, and retrospective gains are not independently confirmed.

e) Implication for research practice.: A small-budget study should distinguish pooled rupee gains from scaled errors, preserve unfavorable groups, disclose dependence assumptions and separate source-attributable execution from independent future confirmation. Historical source audits additionally motivate checking coverage before attributing error changes to market drift. These are practical evaluation lessons, not causal findings or new statistical theory. The value of this narrower contribution must be judged against the target venue's expectations, rather than inferred from dataset scale or a model label.

III. Retained data and fixed-panel scope

The primary experiment uses retained silver v2 mandi reports, combining the AGMARKNET Kaggle archive before October 2025 with official data.gov.in reports thereafter. Prices are INR per quintal. Input Parquet bytes and shard lists are hashed before and after feature execution; these retained bytes, rather than an assurance of identical later API responses, define the reproduced dataset. Official resource access requires a key, which is excluded from source and archives.

The 36-commodity panel is an existing table of 11,845 eligible market–commodity–variety series, supplied as a fixed, hashed external input. Its original selection is not reproduced or independently certified here. Depending on horizon and cleaning regime, the evaluation contains 4,602–5,182 series, rather than all panel members. Findings apply to these eligible report-days and cannot represent every Indian market or farmer. Both cleaning variants are newly built from the same inputs, with history beginning 2022-10-01, for warm-up before 2024 training. The cleaned variant removes reports flagged by a fivefold deviation from the prior ten-report geometric mean; the unfiltered variant retains them. Other validity checks, harmonisation, panel and build settings are held fixed. This does not create manual error labels: unusual prices can be genuine. Common origins may still carry different targets and features after filtering, so the comparison is pipeline sensitivity, not a causal effect of cleaning.

Build records bind the committed source, executed assembly JAR, commands, runtime versions and exit codes to output checksums. This is observed execution attestation, not an independent build witness. Large inputs are retained locally outside Git; a public immutable archival package and independent-environment reproduction remain outstanding. The earlier national-scale source audit and full-size experiments are historical supporting material in the historical supplement, not acceptance-critical validation of this primary study.

IV. Targets, chronological design and metrics

a) Targets.: For entity e, origin date t and modal price $P_{e,t}$, predict the log return to the first report in $[t+h, t+h+3]$ for $h = 3, 7, 14$ calendar days. The next-report target is the first subsequent report within seven days. If no report exists, that origin/horizon is unlabelled, not imputed. The numerical forecast is $P_{e,t}\exp(\hat{y})$. All price information used for features is dated at or before the origin; same-day cross-market reporting availability is an explicit assumption, not established intra-day availability.

b) Periods and budget.: Training origins and targets lie in 2024; 500,000 deterministic hash-selected rows per horizon train each variant. January–June 2025 selects leaf count and early-stopping iteration; July–December 2025 calibrates interval residuals and volatility bins. Retrospective evaluation runs from 2026-01-01 through 2026-10-22. Target dates are purged at each period end. Category encoding and training scales use training rows only. Both variants use seed 20261023, leaf counts 31/63, at most 800 boosting iterations and early stopping after 75 rounds. Selection observations are not reused for calibration. However, all 2026 outcomes were already available during study design, so this separation is not independent confirmation or a pristine future test.

c) Different error questions.: MAE is the mean absolute rupee error across eligible evaluation rows, and relative MAE divides candidate MAE by contemporaneous persistence MAE on identical rows. Frequently reported and high-error series can dominate this pooled measure. For entity e and horizon h, MASE divides each absolute error by that entity's mean absolute naive h-target error over all eligible *training* rows, and then averages over rows with positive finite scales [6]. It is not division by evaluation-period persistence MAE and is not equal weighting of markets. Rows without a usable training scale are excluded only from MASE (4.4–4.5% here), not from full-row MAE. Therefore a relative MAE below one can coexist with MASE near or above one. Neither establishes universal superiority. We also retain RMSE, WAPE, bias, subgroup errors, entity-ratio quantiles, interval coverage and width in the artifacts. Coverage of a nominal price interval is the observed fraction of realised prices inside it, distinct from uncertainty about an aggregate MAE ratio.

d) Dependence sensitivity.: Temporal analysis aggregates all same-day cross-market losses and counts. There are 281–294 calendar days, not half a million independent replicates. We retain all circular/non-circular moving-block specifications of 28, 42 and 56 days, with 2,000 replicates each; endpoints are approximate percentile sensitivity estimates, not validated coverage guarantees. A 56-day span gives only about five non-overlapping spans, not additional independent seasons. Block choice and dependence assumptions matter [33]. Secondary asymptotic normal HAC diagnostics use bandwidths 28/42/56 without a fixed-horizon HLN correction. Each bandwidth has its own Holm-adjusted eight-comparison family; no favourable specification is selected. These choices followed inspection of earlier results and are explicitly retrospective. Original heuristic p values are retained only as legacy diagnostics in artifacts.

Daily loss/count and autocorrelation diagnostics, commodity/state summaries and exploratory entity resampling describe sensitivity. Entities sharing markets, regions and commodity shocks are not independent markets. None of these summaries establishes stationarity, two-way inferential coverage, training uncertainty or future transportability. A human statistical review remains outstanding.

V. Methods

a) Features.: Each origin carries 24 base features plus one horizon-specific seasonal feature, all computed from reports dated $\leq t$ (Table I). Missing numeric values are set to zero; reporting counts and seasonal availability indicators provide partial missingness context. The commodity vocabulary is fitted on training rows. The cross-market features use other markets' reports of the *same day*, which assumes those are published by the end of day t. The older historical study examined previous-day cross-market features; that ablation is not rerun in this bounded experiment.

b) Baselines and learner.: The five baselines are persistence, 7/30-day geometric rolling averages, seasonal naive with persistence fallback, and seasonal-adjusted persistence using last year's horizon change. They are evaluated on the same eligible rows as the candidate. SES, MLlib and three-seed LightGBM comparisons belong to the older historical study and are not rerun or claimed independently confirmed here. Each horizon/variant trains a LightGBM L1 log-return model with learning rate 0.05, minimum leaf size 200, feature fraction 0.8 and bagging fraction 0.7. Early stopping uses selection-period log-return L1 loss; leaf count is selected by rupee MAE on that period. One seed and a deliberately small search budget bound this study; they do not establish best possible predictive performance or training stability.

c) Price intervals.: Only calibration-period residuals and 30-day log-price volatility define the five volatility bins and residual quantiles. The 5/10/90/95% quantiles produce nominal 80/90% price intervals. This implements a baseline residual calibration, not new conformal methodology. Temporal dependence and changing distributions prevent an exchangeability-based coverage guarantee [27]; coverage is measured retrospectively, not promised for future prices.

VI. Primary clean-source study results

a) Clean-source release-study results.: This is a new end-to-end execution of the smaller study, with identical historical periods and model search budget; it is not an exact replication of the full-size historical experiment. The pre-build source tag is `agdata-research-source-v1`. Both feature tables were rebuilt, with source/input/executable snapshots checked before and after execution. The fixed historical panel is an explicitly hashed external input, not a reproduced panel-selection process. Source attribution is stronger than in the earlier runs, but an observed execution log is not an independent build witness. All results remain retrospective and conditional

TABLE I
FEATURE GROUPS ($\ell = \ln P$); ALL FRAMES END AT THE ORIGIN.

Group	Features
Level and lags	ℓ_t; differences to the 1st, 2nd and 5th previous reports; days since the previous report
Rolling (7/30/90 d)	deviation from the rolling mean; standard deviation (30, 90); report counts (30, 90); distance to the 30-day max and min
Range	$(\max - \min)/\text{modal}$ on day t
Seasonal	deviation from the level at $t - 365 \pm 3$; availability indicator; last year's change from $t - 365$ to $t + h - 365$
Cross-market	same-day mean 30-day deviation and last return of the commodity, nationally and within the state; number of reporting markets
Calendar	day of year (sine, cosine); day of week
Categorical	commodity

on the fitted models and observed panel. Across the eight fits, observed pooled MAE reductions are 4.4–5.9%, or INR 9.92–18.83 per quintal. Table II gives absolute errors and exposure. These gains are descriptive historical differences, not evidence of profitable farmer decisions or independent LightGBM superiority over Spark models. MASE excludes 4.4–4.5% of rows lacking positive finite training scales, and therefore describes a different population from full-row MAE.

The sensitivity interval endpoints across all six specifications and eight fits span 0.936–0.962; this range is not a combined confidence interval. State and commodity summaries contain 46 group-by-fit comparisons with relative MAE above one; all subgroup counts and errors are retained, without selecting favourable subgroups. Groups and fits overlap, so this count is descriptive, not a multiplicity-adjusted finding.

The daily diagnostics in Figure 1 display loss differences, exposure and autocorrelation for the cleaned 7-day fit; diagnostics for all fits are recorded with the release. Substantial long-range correlation, seasonal changes or exposure variation can invalidate short-memory approximations. HAC bandwidth sensitivity is secondary: no horizon correction is claimed and tiny p values are not evidence of validated inference. Independent entity-resampling intervals are exploratory and cannot establish generalisation to independent new markets. Human expert statistical review and genuinely future-issued evaluation remain outstanding.

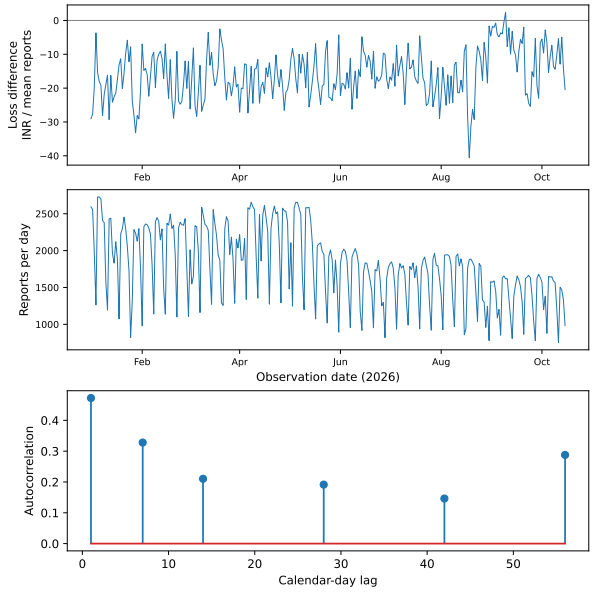


Fig. 1. Release cleaned 7-day daily candidate-minus-persistence total loss divided by mean daily exposure (top), report counts (middle), and descriptive autocorrelation (bottom). No independence threshold is asserted.

VII. PRACTICAL INTERPRETATION OF SCALED ERRORS AND INTERVALS

The primary candidate has MASE 1.004–1.110, despite relative MAE below one against contemporaneous persistence (Table V). These measures ask different questions: MASE uses the entity's training-period naive scale, whereas relative MAE compares two evaluation-period forecasts on the same rows. The denominators, aggregation and even retained row populations differ. A lower pooled rupee MAE therefore supports a modest historical average gain, not a universal victory over all benchmarks, every market or the training-period scale. MASE exclusions (4.4–4.5%) and subgroup failures remain part of the result, not inconvenient observations to omit.

Observed price-interval coverage is 77.7–79.5%, compared with nominal 80%; undercoverage is most evident at the next report. This measured historical frequency does not validate future coverage under dependence or changing conditions. Wider intervals do not guarantee an economically useful decision. These price intervals must also be distinguished from block-bootstrap sensitivity intervals for the aggregate MAE ratio: the latter describe conditional resampling variability, not ranges for a future mandi price. Neither is a profit guarantee.

VIII. HISTORICAL CONTEXT AND FUTURE-ISSUED EVALUATION

a) Supporting historical context.: The separate historical supplement preserves earlier national-scale MLlib benchmarks, source-completeness comparisons, timing ablations, refits and decision-rule backtests. Their original full-size execution and panel selection have not been reproduced end-to-end. They are supporting history, not the primary contribution or validation of this bounded study. The historical matched-origin mirror/official comparison motivates source audits before interpreting drift; it does not isolate causal contributions of changed features, target reports or missingness.

TABLE II
CLEAN-SOURCE RELEASE STUDY: CANDIDATE/PERSISTENCE MAE IN INR PER QUINTAL, FORECAST ROWS, CALENDAR DAYS AND MASE-EXCLUDED ROWS. FORECAST ROWS ARE NOT INDEPENDENT REPLICATES.

Regime	Horizon	Rows	Days	MAE C/P	Rel. MAE	MASE omitted
cleaned	next	553,793	294	192.34/202.26	0.951	24,276
cleaned	h3	531,316	292	238.61/252.95	0.943	24,008
cleaned	h7	519,244	288	301.36/317.31	0.950	22,739
cleaned	h14	496,298	281	393.48/411.51	0.956	22,305
unfiltered	next	554,112	294	197.04/207.30	0.951	24,292
unfiltered	h3	531,606	292	243.52/258.90	0.941	24,024
unfiltered	h7	519,527	288	307.13/323.36	0.950	22,756
unfiltered	h14	496,581	281	397.39/416.21	0.955	22,321

TABLE III
CIRCULAR-BLOCK RELATIVE-MAE PERCENTILE SENSITIVITY INTERVALS (2,000 REPLICATES, NOMINAL 95%). ACTUAL COVERAGE IS UNVALIDATED; ALL BLOCK SPECIFICATIONS ARE SHOWN.

Regime	Horizon	28 days	42 days	56 days
cleaned	next	[0.947, 0.955]	[0.947, 0.955]	[0.947, 0.955]
cleaned	h3	[0.940, 0.948]	[0.940, 0.947]	[0.940, 0.947]
cleaned	h7	[0.946, 0.953]	[0.946, 0.953]	[0.946, 0.953]
cleaned	h14	[0.952, 0.961]	[0.953, 0.961]	[0.953, 0.960]
unfiltered	next	[0.945, 0.955]	[0.946, 0.955]	[0.946, 0.955]
unfiltered	h3	[0.936, 0.946]	[0.936, 0.945]	[0.936, 0.945]
unfiltered	h7	[0.946, 0.954]	[0.946, 0.954]	[0.946, 0.954]
unfiltered	h14	[0.949, 0.960]	[0.950, 0.959]	[0.952, 0.958]

TABLE IV
NON-CIRCULAR-BLOCK SENSITIVITY INTERVALS UNDER THE SAME SETTINGS. LONG BLOCKS LEAVE FEW TEMPORAL SPANS AND DO NOT ESTABLISH STATIONARITY OR FUTURE TRANSPORTABILITY.

Regime	Horizon	28 days	42 days	56 days
cleaned	next	[0.946, 0.955]	[0.946, 0.955]	[0.947, 0.955]
cleaned	h3	[0.939, 0.947]	[0.939, 0.947]	[0.940, 0.947]
cleaned	h7	[0.946, 0.954]	[0.946, 0.953]	[0.946, 0.952]
cleaned	h14	[0.952, 0.962]	[0.952, 0.961]	[0.953, 0.959]
unfiltered	next	[0.946, 0.956]	[0.946, 0.955]	[0.947, 0.955]
unfiltered	h3	[0.938, 0.946]	[0.939, 0.946]	[0.939, 0.946]
unfiltered	h7	[0.946, 0.955]	[0.947, 0.955]	[0.947, 0.954]
unfiltered	h14	[0.951, 0.961]	[0.952, 0.960]	[0.953, 0.958]

TABLE V
PRIMARY STUDY SCALED ERRORS AND OBSERVED NOMINAL 80% PRICE-INTERVAL PERFORMANCE. C/P DENOTE CANDIDATE/PERSISTENCE. WIDTH IS DIVIDED BY ORIGIN PRICE. MASE EXCLUDES UNUSABLE TRAINING SCALES.

Regime	Horizon	MASE C	MASE P	Coverage 80%	Rel. width
cleaned	next	1.100	1.156	77.7%	0.217
cleaned	h3	1.004	1.060	79.1%	0.272
cleaned	h7	1.074	1.109	79.4%	0.338
cleaned	h14	1.044	1.061	79.4%	0.442
unfiltered	next	1.110	1.168	77.8%	0.219
unfiltered	h3	1.014	1.071	79.1%	0.273
unfiltered	h7	1.080	1.117	79.5%	0.342
unfiltered	h14	1.048	1.065	79.5%	0.443

b) Exploratory adoption boundary.: The prototype's LightGBM serving choice used a period already examined during development. Neither this smaller experiment nor its cleaning variants independently confirms LightGBM superiority over newly refitted Spark alternatives. The primary question here is its observed performance against the five specified simple baselines under a recorded, bounded execution.

c) Future collection, not completed confirmation.: A new local issuance utility copies and hashes the existing prototype boosters, feature/category metadata and calibration, records source and dependencies, and recomputes forecasts using those frozen artifacts. Only recent prior-day origins whose entire

target window starts strictly after the local issuance date qualify. The CLI accepts no backdating override; feature-file hashes, future-date rejection, duplicate keys and a chained batch manifest support integrity checks. This future-window population can exclude next-report forecasts and differs from the retrospective-origin population, which must be reported rather than silently equated. The frozen prototype is not the smaller release-study model, so this collection cannot directly confirm that model's future gain. The local system clock and hash chain are not external time attestation or proof of point-in-time raw-report availability. No outcomes are scored or future success assumed. At least 90 completed calendar days plus target maturation, a reviewed scoring protocol, independent archival and human statistical review remain required before any confirmatory interpretation. No issuance schedule is installed. The operational protocol and actual collection status are kept outside retrospective result tables.

IX. Discussion, limitations and practical meaning

The primary evidence is a modest reduction of pooled rupee error under two cleaning pipelines in one already observed evaluation period. MASE near or above one, omitted scales and worse-performing subgroups constrain its interpretation. Some error gains may matter operationally, but modal price is a published market summary, not a farmer's transaction price; transport, fees, perishability and storage can outweigh these differences. No profit or welfare effect is estimated [31], [32].

The statistical sample includes 281–294 calendar days, not hundreds of thousands of independent trials. Shared shocks, overlapping targets, changing report counts and long-range autocorrelation complicate inference. A 56-day block yields only about five non-overlapping temporal spans; 2,000 resamples cannot create independent seasons. The sensitivity grid is useful disclosure, not validated coverage or a substitute for human statistical review. Independent entity resampling does not make correlated markets independent. Training-sample and hyperparameter-search uncertainty are not captured.

Other limitations are the fixed historical panel, one training year, one seed, two leaf settings, source revisions, irregular target availability, omitted weather/arrivals, and assumed end-of-day availability of same-day cross-market features. The evaluation year was visible during design. The clean execution is not independent confirmation or a reconstruction of the older national-scale benchmark. We do not extend the primary evaluation into the selection/calibration periods merely to obtain more observations: doing so would change their roles and would not add independent seasons. Longer independent observation requires time-forward collection or a separately defined rolling-origin study.

The main reproducibility claim is scoped to this smaller retained-input execution. A public immutable archive of exact large input bytes, a separate-environment run, and a human statistician's assessment remain outstanding. IJF is the intended empirical forecast-evaluation venue, not a claim of readiness or acceptance. We do not propose a new learner, new conformal theory, causal missingness effect or superiority over papers with incompatible data and protocols. The six-language prototype and historical decision cards are supporting system work, not experimentally demonstrated farmer benefit; native-speaker, device and farmer reviews remain pending.

X. Conclusions and reproducibility boundary

A bounded, clean-source retrospective study shows 4.4–5.9% lower pooled MAE than contemporaneous persistence across four horizons and two cleaning regimes. MASE remains approximately 1.00–1.11 using training-period scales, with 4.4–4.5% of rows omitted only from that metric. The conclusion is modest historical average error reduction under specified settings, not universal predictive superiority or improved farmer profit.

The contribution is a reproducible small-budget cleaning/dependence sensitivity study on retained inputs. Historical national-scale comparisons and the source-completeness audit are separated as supporting material; their original full-size execution and panel selection are not certified by this work. Dependence sensitivities remain approximate and retrospective. Frozen-model future issuance can begin collecting a different, explicitly defined population, but no future confirmation or human statistical approval is claimed.

a) Artifacts and availability.: Fitting source is tagged `agdata-research-source-v1`; release records bind pre/post source/input/JAR hashes, commands and feature outputs. Saved predictions and generated primary tables are recorded under `results/release_v1`. Models, large data and issuance batches remain local outside Git. Public source/results tags and compact manifests do not constitute a public immutable data archive, independently witnessed execution or anonymised journal submission package. Later API responses can differ from retained bytes after revisions. The fixed panel is a declared external input. Corresponding-author details, declarations and formal journal-template compliance remain unverified.

The evidence ledger, commands, third-review response and future protocol distinguish reproduced retrospective results, historical supporting evidence and pending forward evaluation. Human expert statistical review, independent archival/reproduction and completed prospective outcomes remain necessary before a stronger submission recommendation.

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References

[1] R. K. Paul, M. Yeasin, P. Kumar, P. Kumar, M. Balasubramanian, H. S. Roy, A. K. Paul, and A. Gupta, "Machine learning techniques for forecasting agricultural prices: A case of brinjal in Odisha, India," *PLOS ONE*, vol. 17, no. 7, p. e0270553, 2022.

[2] R. L. Manogna, V. Dharmaji, and S. Sarang, "Enhancing agricultural commodity price forecasting with deep learning," *Scientific Reports*, vol. 15, no. 1, p. 20903, 2025.

[3] P. Kumar, P. S. Badal, R. K. Paul, G. K. Jha, P. Venkatesh, Kamalvanshi, P. Anbukani, M. Balasubramanian, and P. Patel, "Forecasting onion price for Varanasi market of Uttar Pradesh, India," *The Indian Journal of Agricultural Sciences*, vol. 91, no. 2, pp. 249–253, 2021.

[4] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and machine learning forecasting methods: Concerns and ways forward," *PLOS ONE*, vol. 13, no. 3, p. e0194889, 2018.

[5] ——, "The M4 competition: 100,000 time series and 61 forecasting methods," *International Journal of Forecasting*, vol. 36, no. 1, pp. 54–74, 2020.

[6] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.

[7] T. Gneiting, "Making and evaluating point forecasts," *Journal of the American Statistical Association*, vol. 106, no. 494, pp. 746–762, 2011.

[8] S. Kolassa, "Evaluating predictive count data distributions in retail sales forecasting," *International Journal of Forecasting*, vol. 32, no. 3, pp. 788–803, 2016.

[9] H. Hewamalage, K. Ackermann, and C. Bergmeir, "Forecast evaluation for data scientists: common pitfalls and best practices," *Data Mining and Knowledge Discovery*, vol. 37, no. 2, pp. 788–832, 2022, published online 2022; appears in volume 37 (2023).

[10] K. O. Bokun, L. E. Jackson, K. L. Kliesen, and M. T. Owyang, "Fred-sd: A real-time database for state-level data with forecasting applications," *International Journal of Forecasting*, vol. 39, no. 1, pp. 279–297, 2023.

[11] R. K. Paul, M. Yeasin, C. Tamilselvi, A. K. Paul, P. Sharma, and P. S. Birthal, "Can deep learning models enhance the accuracy of agricultural price forecasting? Insights from India," *Intelligent Systems in Accounting, Finance and Management*, vol. 32, no. 1, p. e70002, 2025.

[12] R. K. Paul and S. Garai, "Performance comparison of wavelets-based machine learning technique for forecasting agricultural commodity prices," *Soft Computing*, vol. 25, no. 20, pp. 12 857–12 873, 2021.

[13] G. K. Jha and K. Sinha, "Time-delay neural networks for time series prediction: an application to the monthly wholesale price of oilseeds in India," *Neural Computing and Applications*, vol. 24, no. 3-4, pp. 563–571, 2012, published online 2012; appears in volume 24 (issue 3–4).

[14] ——, "Agricultural price forecasting using neural network model: An innovative information delivery system," *Agricultural Economics Research Review*, vol. 26, no. 2, pp. 229–239, 2013. [Online]. Available: http://ageconsearch.umn.edu/record/162150

[15] P. Anjoy, R. K. Paul, K. Sinha, A. K. Paul, and M. Ray, "A hybrid wavelet based neural networks model for predicting monthly WPI of pulses in India," *The Indian Journal of Agricultural Sciences*, vol. 87, no. 6, pp. 834–839, 2017.

[16] A. Praveenkumar, G. K. Jha, S. D. Madival, A. Lama, and R. R. Kumar, "Deep learning approaches for potato price forecasting: Comparative analysis of LSTM, Bi-LSTM, and AM-LSTM models," *Potato Research*, vol. 68, no. 2, pp. 1941–1963, 2024, published online 2024; appears in volume 68.

[17] R. Jaiswal, G. K. Jha, R. R. Kumar, and K. Choudhary, "Deep long short-term memory based model for agricultural price forecasting," *Neural Computing and Applications*, vol. 34, no. 6, pp. 4661–4676, 2021, published online 2021; appears in volume 34.

[18] J. Wang, Z. Wang, X. Li, and H. Zhou, "Artificial bee colony-based combination approach to forecasting agricultural commodity prices," *International Journal of Forecasting*, vol. 38, no. 1, pp. 21–34, 2022.

[19] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions," *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.

[20] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.

[21] C. Bergmeir, R. J. Hyndman, and B. Koo, "A note on the validity of cross-validation for evaluating autoregressive time series prediction," *Computational Statistics & Data Analysis*, vol. 120, pp. 70–83, 2018.

[22] V. Cerqueira, L. Torgo, and I. Mozetič, "Evaluating time series forecasting models: an empirical study on performance estimation methods," *Machine Learning*, vol. 109, no. 11, pp. 1997–2028, 2020.

[23] A. Jain, S. Marvaniya, S. Godbole, and V. Munigala, "A framework for crop price forecasting in emerging economies by analyzing the quality of time-series data," arXiv:2009.04171, 2020.

[24] H. Guo, A. Woodruff, and A. Yadav, "Improving lives of indebted farmers using deep learning: Predicting agricultural produce prices using convolutional neural networks," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 08, pp. 13 294–13 299, 2020.

[25] C. Xu and Y. Xie, "Conformal prediction for time series," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 10, pp. 11 575–11 587, 2023.

[26] V. Jensen, F. M. Bianchi, and S. N. Anfinsen, "Ensemble conformalized quantile regression for probabilistic time series forecasting," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 7, pp. 9014–9025, 2024.

[27] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021, preprint: arXiv:2106.00170, doi:10.48550/arXiv.2106.00170. [Online]. Available: https://proceedings.neurips.cc/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html

[28] J. Ouyang and C. Tang, "Drift-aware conformal prediction for reliable potato price forecasting under nonstationary markets," *Potato Research*, vol. 69, no. 3, 2026.

[29] R. Jensen, "The digital provide: Information (technology), market performance, and welfare in the South Indian fisheries sector," *The Quarterly Journal of Economics*, vol. 122, no. 3, pp. 879–924, 2007.

[30] A. Goyal, "Information, direct access to farmers, and rural market performance in Central India," *American Economic Journal: Applied Economics*, vol. 2, no. 3, pp. 22–45, 2010.

[31] M. Fafchamps and B. Minten, "Impact of SMS-based agricultural information on Indian farmers," *The World Bank Economic Review*, vol. 26, no. 3, pp. 383–414, 2012.

[32] S. Mitra, D. Mookherjee, M. Torero, and S. Visaria, "Asymmetric information and middleman margins: An experiment with Indian potato farmers," *The Review of Economics and Statistics*, vol. 100, no. 1, pp. 1–13, 2018.

[33] D. N. Politis and H. White, "Automatic block-length selection for the dependent bootstrap," *Econometric Reviews*, vol. 23, no. 1, pp. 53–70, 2004.